IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

MICROELECTROMECHANICAL SYSTEMS (MEMS)

Friday 19 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer Question 1.
Answer any TWO of the remaining questions (Questions 2, 3 and 4).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : A. Holmes
Second Marker(s) : Z. Durrani

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Information for Candidates

Permittivity of free space: $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

The bending equation for the deflection $w(x, y)$ of a plate subject to a normal load $p(x, y)$ is:

$$\nabla^2(\nabla^2 w) = \frac{p}{D}$$

where $D = \frac{Eh^3}{12(1-\upsilon^2)}$ is the flexural rigidity.

The 2D Laplacian operator in polar coordinates (r, θ) is:

$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r} r \frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}$$

The generalised Hooke's law equations are:

$$E\varepsilon_x = \sigma_x - \upsilon\sigma_y - \upsilon\sigma_z$$
$$E\varepsilon_y = \sigma_y - \upsilon\sigma_z - \upsilon\sigma_x$$
$$E\varepsilon_z = \sigma_z - \upsilon\sigma_x - \upsilon\sigma_y$$

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This question is compulsory

1. a) By considering the electrostatic force developed by a parallel plate electrostatic actuator, or otherwise, obtain a scaling law for the electrostatic force assuming constant maximum electric field. Comment on the validity of the constant field assumption at different length scales. [6]

b) Draw a diagram showing the forces and couples acting on a flexure that is subject to transverse end loads. Write down expressions for the variations with position of the bending moment and the elastic strain inside the flexure. Hence show that the maximum strain is:

$$\varepsilon_{max} = \frac{3v_L h}{l^2}$$

where l, h and v_L are the length, width (in the deflection axis) and end deflection of the flexure. [10]

c) Starting from the generalised form of Hooke's law, and applying appropriate constraints to the stress and strain components, derive an expression for the biaxial modulus of a material in terms of its Young's modulus and Poisson's ratio. Give an example a situation where the biaxial modulus would be used in analysis of a MEMS device. [6]

d) By considering the unit cell of silicon, show that the {111} planes form an angle of 54.7° with the surface of a <100>-oriented wafer. Also show that certain {111} planes are perpendicular to the surface of a <110>-oriented wafer. If such a wafer is anisotropically etched, what will be the angle between the vertical sidewalls of an etch-limited cavity? [6]

e) List the main actuation mechanisms used in MEMS devices, and briefly compare them in terms of speed, force and ease of implementation. [6]

f) Describe briefly the principle of piezoelectric transduction. Also show, with the aid of a sketch, how this transduction mechanism might be applied to a cantilever-based silicon micromachined accelerometer. [6]

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Answer ANY TWO of the remaining questions (Questions 2, 3 and 4)

2. Figure 2.1 shows a silicon micromachined accelerometer with capacitive readout. The comb electrode arrays are not drawn to scale, and in the actual device there are $N = 50$ electrode fingers protruding from the mass. The mechanical layer thickness is $h = 10$ μm, the nominal inter-electrode gap is $g = 1$ μm, and the electrode overlap length is $l = 100$ μm. The proof mass has an area of 0.12 mm^2. The suspension consists of four folded springs in which the flexures all have length 250 μm and width 5 μm.

   a) Calculate the suspension stiffness and the resonant frequency of the device, assuming E = 169 GPa for silicon. The density of silicon is 2330 kg/m^3. [8]

   b) Using a simple parallel-plate model, and ignoring parasitics, estimate the capacitance C_A between terminal A and ground when the mass is in its equilibrium position at zero applied acceleration. Would you expect your estimate to be lower or higher than the actual value? Explain your reasoning. [8]

   c) Derive a general expression for differential capacitance $C_{diff} = (C_A - C_B)$ as a function of proof mass displacement. Hence determine the magnitude of C_{diff} when the accelerometer is operated in open loop and subject to an acceleration of 10 ms^{-2}. [8]

   d) Sketch the block diagram of a suitable readout circuit for the accelerometer and explain briefly how the circuit works. [6]

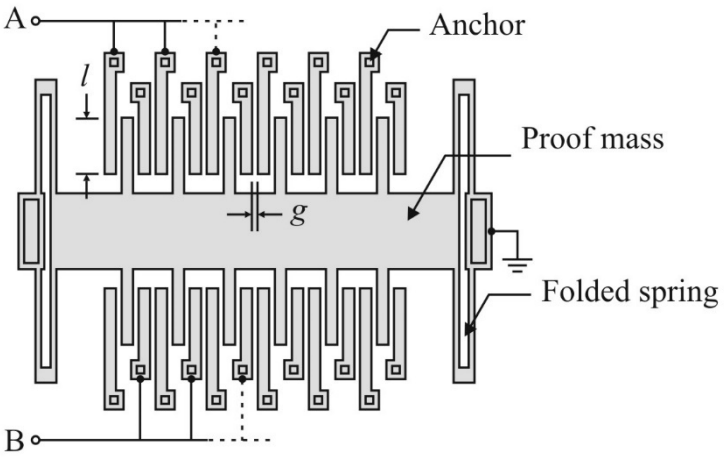


Figure 2.1

3. a) Show that the function $w(r)$ below satisfies the bending equation for a circular plate of radius a that is built in around the periphery and subject to a uniform normal load. Also give an expression for the parameter w_0 in terms of the applied load and the radius and flexural rigidity of the plate.

$$w(r) = w_0(1 - r^2/a^2)^2$$

[10]

b) Figure 3.1 shows a capacitive pressure sensor with a circular membrane, fabricated in a BSOI (bonded silicon on insulator) wafer with two mechanical layers. The thickness of each mechanical layer is h = 10 µm and the oxide layer between the two mechanical layers is g = 2 µm thick. The recess in the handle wafer is cylindrical with radius a = 0.5 mm.

i) Suggest a possible fabrication sequence for the device, starting from the unprocessed BSOI wafer. You should specify the process used at each step of the sequence, but you do not need to describe the processes in detail.

[6]

ii) Calculate the device capacitance, C, neglecting the perforations in the top electrode and the parasitic capacitance due to the support around the periphery. [3]

iii) Using the result in part a), show that, if $w \ll g$, the shift ΔC in capacitance when a differential pressure p is applied can be expressed approximately as:

$$\Delta C \approx -\frac{\pi}{16}\frac{\varepsilon_0 a^6}{g^2 h^3}\frac{1-\nu^2}{E}p$$

(hint: if the plate deflection is small then you can assume the local change in the capacitance per unit area is approximately $\frac{d}{dg}\left(\frac{\varepsilon_0}{g}\right) \times w$.) [8]

iv) Calculate the shift in capacitance when the sensor is subject to a differential pressure of 1 kPa. Assume values of 160 GPa and 0.22 for the Young's modulus on Poisson ratio of silicon. [3]

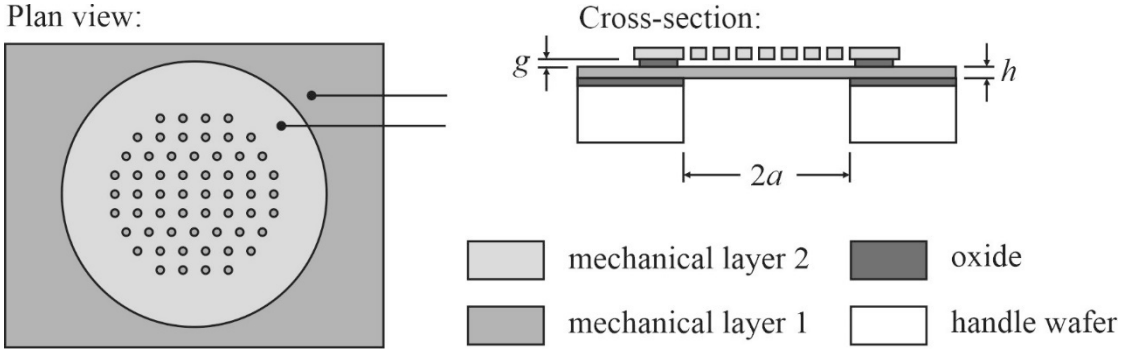


Figure 3.1

4. a) Write down expressions for the transverse stiffness, k_T, and the axial stiffness, k_A, of a surface micromachined flexure. Hence show that the ratio of transverse to axial stiffness is:

$$\frac{k_T}{k_A} = \alpha = \left(\frac{w}{l}\right)^2$$

where *w* is the width of the beam and *l* is its length. [8]

b) Explain briefly why electrothermal actuators generally make use of frustrated thermal expansion as opposed to relying on unconstrained thermal expansion. [4]

c) Figure 4.1 shows a so-called V-beam electrothermal actuator which comprises two flexures connected in series in a chevron configuration, i.e. with an angle between them. Figure 4.1a shows the actuator both in its rest state and in its deflected state when subject to Joule heating. Figure 4.1b shows the axial and transvers loads and deflections for the right-hand flexure in the deflected state.

Explain why the loads and deflections in Figure 4.1b must satisfy the following relations:

$$P_A \sin\theta = P_T \cos\theta \quad ; \quad u_A \cos\theta = u_T \sin\theta$$ [4]

Also write down equations relating the loads P_A, P_T to the deflections u_A, u_T, taking into account the contribution of the thermal strain, ε_{th}, to u_A. Hence show that the deflection Δy of the actuator may be written as:

$$\Delta y = l\varepsilon_{th}\frac{\sqrt{1+u^2}}{u+\alpha/u}$$

where $u = \tan\theta$. [8]

d) Using the result in part c), show that, for a given thermal strain, the deflection will be maximised in an actuator for which $\theta \approx w/l$. You may assume that $w/l \ll 1$. [6]

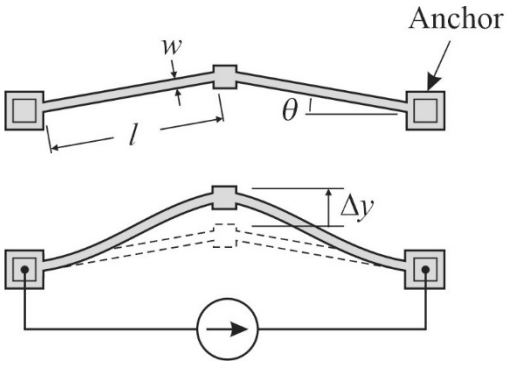


Figure 4.1a

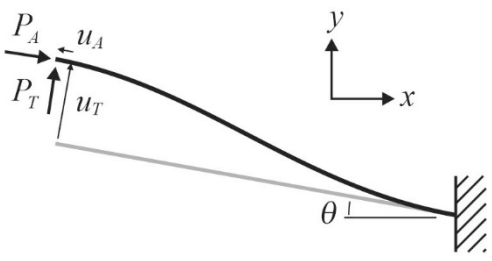


Figure 4.1b

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